Badreddine Benfettouma

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Research interests

Digital integrated-circuit design and RISC-V microarchitecture; formal verification of hardware; energy-efficient hardware for machine-learning inference; open-source EDA flows and open PDKs.

Education

Licence 3 (third year of the Bachelor), Electronics 2026–2027
Université Bourgogne Europe, Dijon, France

Licence 1 and 2, Computer Science and Electronics track 2024–2026
Université de Bourgogne (Université Bourgogne Europe since January 2025), Dijon, France
Selected grades (out of 20): computer architecture 14.2 (lab 18.1), signal processing 17.5, scientific computing 20.

The French *Licence* is a three-year bachelor's degree (180 ECTS).

Projects

RV32I-Alpha: a two-stage RISC-V processor core 2026
VHDL, SystemVerilog; source code

- Two-stage pipelined core implementing the 37 instructions of the RV32I base ISA, with traps on illegal instructions and misaligned accesses and a data-memory port that accepts wait states. Written in VHDL, then ported to SystemVerilog.
- Formal verification: riscv-formal instruction and consistency checks (bounded model checking, 10–15 cycles) and custom SymbiYosys properties for the memory ports; mutation testing (8 injected bugs, all detected); Yosys proof that the VHDL and SystemVerilog versions are sequentially equivalent.
- FPGA: Digilent Basys 3 board (Artix-7); the core alone uses 675 LUTs and meets 100 MHz (Vivado).
- ASIC: hardened to a DRC- and LVS-clean GDSII macro in SkyWater SKY130 with LibreLane (power grid, clock tree, sign-off; 0.22 mm^2, 40 MHz at the typical corner).

Area and energy accounting of logic-gate neural networks 2026
Independent project, write-up in progress; Python, PyTorch, Yosys, OpenROAD, OpenSTA

- Built an open flow that takes a trained logic-gate network to a placed SKY130 netlist, simulates it at gate level on real test images and derives power from the recorded switching activity.
- Checked that the gate-level netlist reproduces the software model bit for bit. Included the cost of input buffering, output logic and wiring, which published comparisons usually leave out.

4-bit weight formats for LLM inference hardware 2026
Independent project; Python, PyTorch, Hugging Face Transformers; source code

- Perplexity of five 4-bit weight formats on three LLMs. At equal quality, trellis coding saves 11–17 % of the bits against NVFP4 but only 6–9 % against a rotated Lloyd-Max format.

Technical skills

HDLs	VHDL, SystemVerilog, Verilog
Verification	SymbiYosys, riscv-formal, Yosys equivalence checking, GHDL, Icarus Verilog
Implementation	Vivado (Artix-7); LibreLane, Yosys, OpenROAD, OpenSTA, Magic, KLayout
Circuits	ngspice with SKY130 device models, Monte-Carlo mismatch analysis
Programming	Python (NumPy, PyTorch, Transformers), Tcl, RISC-V assembly; Linux, Git

Languages

French	native
English	C1
Japanese	learning; level certification in progress
Arabic	Algerian Arabic (spoken); reads the Arabic script

October 2026